

Numerical Protocol

Log-Volatility Roughness Evaluation
under the Log-Normal SV DGP with fBM

Othmane Zarhali

Abstract

We evaluate the roughness of $\log V_t$ under the log-normal Stochastic volatility DGP:

1. **Data generation.** Simulate the log-normal stochastic volatility process

$$\sigma_t = \sigma_0 \exp\left(\lambda B^H(t) - \frac{\lambda^2}{2} t^{2H}\right),$$

where the fractional Brownian motion B^H is generated by Cholesky decomposition of its covariance matrix.

2. **Reference properties.** Use the analytical properties of the DGP as the benchmark:
 - the variogram has an exact slope equal to $2H$;
 - the log volatility process is monofractal, satisfying

$$\zeta(q) = qH, \quad H(q) \equiv H, \quad \mu(q) \equiv 0.$$

3. **Model calibration.** Calibrate three competing models to the complete stylised-fact profile of the DGP:
 - ESN A2-981003, using its current optimal architecture

$$(N_r, N_z) = (96, 12),$$

and the smooth Gaussian-kernel score;

- QRH;
 - PDV-GL.
4. **Model evaluation.** Assess each calibrated model by comparing its reproduction of the DGP roughness through three complementary diagnostics:
 - (a) OLS estimation of the Hurst exponent;
 - (b) linearity of the autocovariance as a function of $\tau^{2\hat{H}}$;
 - (c) estimation of the multifractal spectrum.

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DGP: Log-Normal SV with fBM

$$r_t = \sigma_t \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 1), \quad \sigma_t = \sigma_0 \exp\left(\lambda B^H(t) - \frac{\lambda^2}{2} t^{2H}\right). \quad (1)$$

The centring term $-\lambda^2 t^{2H}/2$ ensures $\mathbb{E}[\sigma_t] = \sigma_0$. Analytical truth: $m_2(\tau) = 4\lambda^2 \tau^{2H}$, so OLS variogram slope = $2H$ exactly. DGP is monofractal: $\zeta(q) = qH$, $H(q) \equiv H$, $\mu(q) \equiv 0$. Simulation via Cholesky of $\Sigma_{ij} = \frac{1}{2}(t_i^{2H} + t_j^{2H} - |t_i - t_j|^{2H})$, cached. Parameter grid available: $H \in \{0.05, 0.10, 0.40\}$, $\lambda \in \{0.5, 1.0, 2.0\}$; this note reports the rough-target subset $H \in \{0.05, 0.10\}$ at $\lambda = 1.0$.

Models

All three models use Gaussian Brownian innovations.

2.1 ESN A2-981003 (current optimal architecture)

7 shared hyperparameters, with $N_r = 96$, $N_z = 12$. **rough_scale**, the z -readout profile (**zr_lo**, **zr_hi**), and the symmetric quadratic matrix coefficients (m_1, m_2) are calibrated per (H, λ) cell, not part of this fixed table (see §3's Calibration table for their per-cell values). A single $\varepsilon_t \sim \mathcal{N}(0, 1)$ drives both the return and the reservoir. H , the r -layer rate window $[\lambda_{lo}, \lambda_{hi}]$, and the z -layer rate window $[a_{lo}, a_{hi}]$ are **calibrated per (H, λ) DGP cell** (§3) – they are not part of the shared architecture above. H_{ESN} (the reservoir's own memory-kernel exponent) is a different object from the DGP's Hurst parameter H_{DGP} ; the inner loop searches over H_{ESN} so the simulated $\hat{H}$ matches the DGP's target, and the two need not coincide.

2.2 QRH

Bourgey & Gatheral (2026): $V_t = Y_t^2 + c$, gamma kernel, Euler-Volterra with RL weights $w_j = \Delta t^\alpha / (\Gamma(\alpha)\alpha)[(j+1)^\alpha - j^\alpha]$, 252-step ring buffer.

2.3 PDV-GL

Guyon & Lekeufack (2023): two-factor power-law kernel, $V_t = (1 - \beta)\bar{V} + \beta[\alpha_1 F_t(\alpha_r) + (1 - \alpha_1)F_t(\alpha_v)]$, online computation.

Calibration

Calibration minimises the full 11-term data-adaptive score S , with the **smooth Gaussian-kernel** proximity:

$$S = \sum_k w_k (1 - f_k) + 5 \cdot \text{Stress}, \quad f(x, c, s) = \exp\left(-\frac{1}{2}\left(\frac{x-c}{s}\right)^2\right) \in (0, 1], \quad (2)$$

$S \geq 0$, $S = 0$ is a perfect match on all 11 stylised facts, $S = 15.3$ is worst-possible; minimised directly (no sign flip). The smooth kernel provides a never-zero gradient at any distance from target, unlike a piecewise-linear tent $\max(0, 1 - |x - c|/s)$, which hits exactly 0 (zero gradient) beyond one tolerance width.

ESN read-out. The ESN’s volatility pre-activation combines three calibrated channels:

$$\eta_t = b_0 + \rho_r \text{rough_scale} (q^\top r_t) + \sum_{j=1}^{N_z} w_{j,z} z_{j,t} + r_t^\top Q r_t, \quad w_{j,z} = \frac{\text{z_readout}_j}{\sqrt{N_z}}, \quad Q = \frac{1}{N_r} (m_1 I + m_2 q q^\top), \quad (3)$$

where $\text{z_readout}_j = \text{geomspace}(\text{zr_lo}, \text{zr_hi}, N_z)[j]$ is a per-mode profile, and Q is a genuine **symmetric** matrix (no positive-definiteness constraint): m_1 scales the reservoir’s isotropic energy, m_2 scales the squared rough factor $(q^\top r_t)^2$ specifically. **Only 7 hyperparameters remain shared** across all (H, λ) cells (**z_strength**, **even_strength**, **linear_strength**, **gamma_norm**, **local_z_strength**, **zz_scale**, **sign_prob_neg**, plus N_r, N_z); **rough_scale**, **zr_lo**, **zr_hi**, and (m_1, m_2) are calibrated per (H, λ) cell, exactly as $H, \lambda_{\text{lo}}, \lambda_{\text{hi}}, a_{\text{lo}}, a_{\text{hi}}$ already were.

Model	Parameters calibrated	Remark
ESN A2-981003	$H, \lambda_{\text{lo}}, \lambda_{\text{hi}}, a_{\text{lo}}, a_{\text{hi}}, \Delta b_0, \text{scale } s, \text{rough_scale}, \text{zr_lo}, \text{zr_hi}, m_1, m_2$	7 shared hyperparameters fixed; full 12-dim. inner-loop calibrated per (H, λ) cell (11-start Nelder–Mead)
QRH	$H, \nu_{\text{vol}}, \lambda, c_{\text{frac}}$	
PDV-GL	$\beta, \alpha_1, \alpha_r, \alpha_v$	

The ESN’s inner loop uses an 11-point multi-start Nelder–Mead scheme (rough/fast, persistent/slow, intermediate, wide-band/multiscale, long-memory extreme, strong-amplitude, weak-amplitude, isotropic-positive, isotropic-negative, q -aligned-positive, and q -aligned-negative starts), keeping the best-of-eleven result.

Per-Cell Calibrated Read-Out Parameters

At $\lambda = 1.0$, the ESN’s calibrated **rough_scale** and z -readout profile stay in a narrow band across the two rough-target H values tested; m_1, m_2 are both near-zero at both cells, with no large excursion in either quadratic coefficient at this narrower H range (this contrasts with a persistent target, $H = 0.40$, tested separately, where m_1 calibrated to a large negative value – consistent with the general caution that this quadratic channel is weakly and inconsistently identified across seeds

and should not be over-interpreted from any single cell).

True H	rough_scale	zr_lo	zr_hi	m_1	m_2	Score S
0.05	0.433	0.032	0.072	-0.0002	-0.0002	3.955
0.10	0.410	0.028	0.061	0.0000	0.0004	4.216

Roughness Diagnostics

5.1 Diagnostic 1: Hurst OLS

$\hat{H} = \frac{1}{2} \cdot \text{OLSslopeslope of } \log m_2(L) \text{ vs } \log L, L \in \{1, 2, 4, 8, 16, 32, 64\}$. Cross-path 95% CI from empirical quantiles.

5.2 Diagnostic 2: ACov linearity

OLS: $C(\tau)/C(0) = a + b\tau^{2\hat{H}}$. $R^2 \rightarrow 1$ = correct Hölder exponent.

5.3 Diagnostic 3: Scale invariance

$S_q(\tau) \sim \tau^{\zeta(q)}$; $H(q) = \zeta(q)/q$; $\mu(q) = \zeta(q) - q\zeta(1)$. Monofractal DGP: all three are constant/zero.

Summary Figure

Cross-diagnostic summary of $\hat{H}$ versus the true H , aggregating Diagnostic 1 across both H values tested.

Monte Carlo Setup

Parameter	Value
DGP paths	30
Cal. paths	4
Cal. days	600
Final paths	30
Final days	2500
Burn-in	500
Δt	1 (daily)
Dyadic lags	$\{1, 2, 4, 8, 16, 32, 64\}$
Structure function q	$\{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$
ACov lags $\tau_{\max}$	20

$\hat{H}$ Estimation — log-normal SV DGP ($\lambda = 1.0$)
Dyadic variogram OLS: slope = $2\hat{H}$

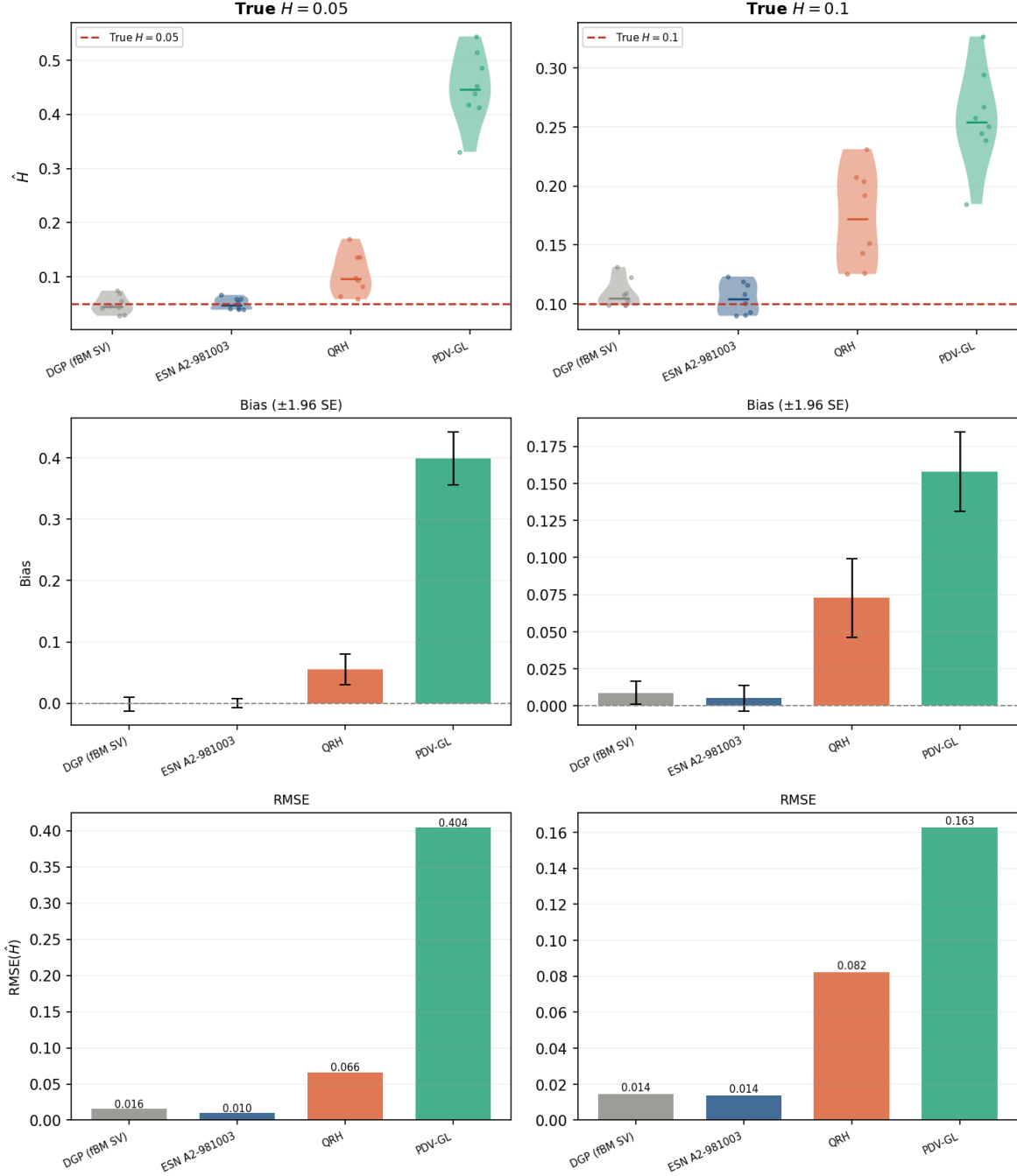


Figure 1: $\hat{H}$ estimation via the dyadic-variogram OLS slope, per model and true $H \in \{0.05, 0.10\}$: violin plots of the cross-path $\hat{H}$ distribution (top row), bias ± 1.96 SE (middle row), and RMSE($\hat{H}$) (bottom row).

ACov of $\log V_t$ vs $au^{2\hat{H}} - (\lambda = 1.0)$
OLS: $C(\tau)/C(0) = a + b\tau^{2\hat{H}} \mid R^2 \rightarrow 1 = \text{correct roughness}$

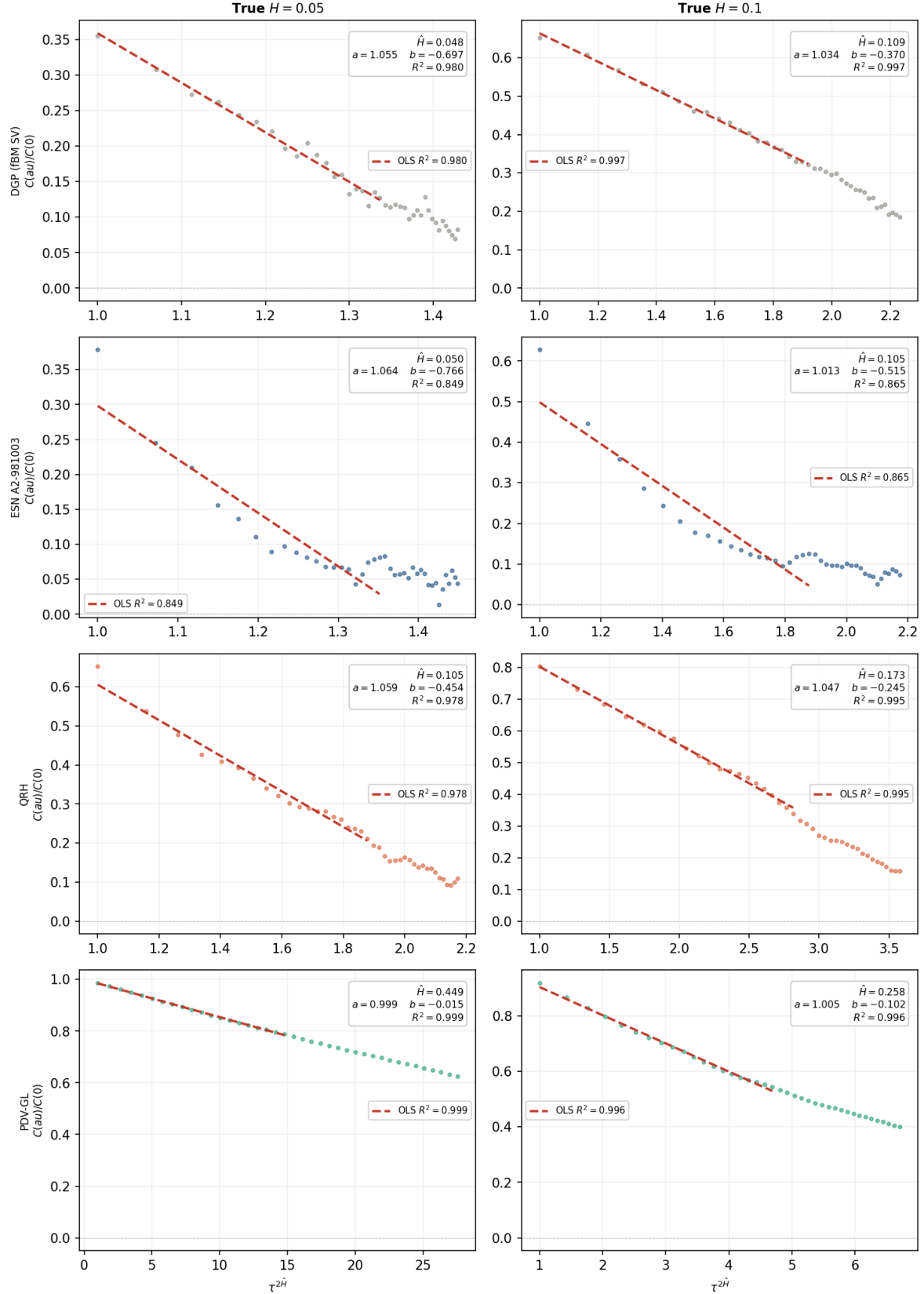


Figure 2: Autocovariance of $\log V_t$ against $\tau^{2\hat{H}}$, per model (rows) and true $H \in \{0.05, 0.10\}$ (columns), with the OLS fit and R^2 annotated in each panel.

Table 1: The 7 shared hyperparameters of the optimal architecture.

Hyperparameter	Value
N_r	96
N_z	12
z_strength	0.1929
even_strength	2.2291
linear_strength	0.1930
gamma_norm	0.9651
local_z_strength	0.0476
zz_scale	0.0397
sign_prob_neg	0.2219
rough_orientation	-1.0 (fixed)

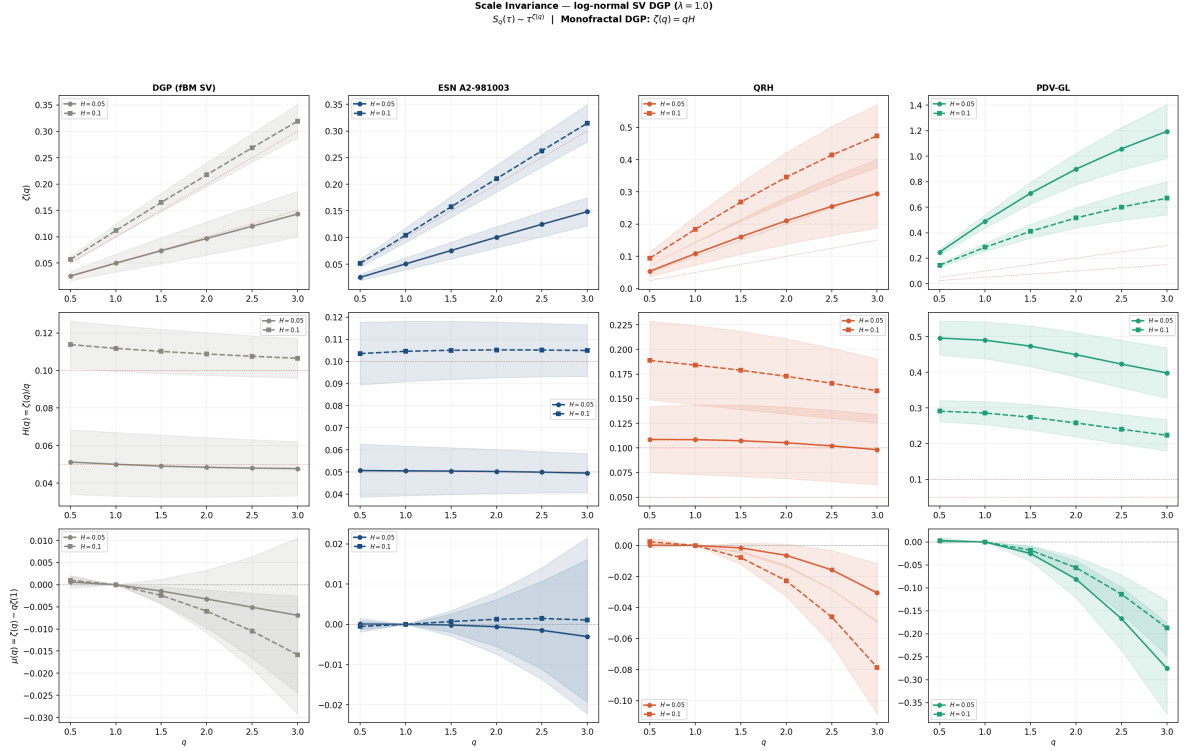


Figure 3: Scale invariance: structure-function exponent $\zeta(q)$ (top row), generalised Hurst function $H(q) = \zeta(q)/q$ (middle row), and multifractality measure $\mu(q) = \zeta(q) - q\zeta(1)$ (bottom row), per model (columns) and true $H \in \{0.05, 0.10\}$ (line style). A flat $H(q)$ and $\mu(q) \equiv 0$ indicate monofractal behaviour, matching the DGP.

Roughness Summary vs True H — ($\lambda = 1.0$)
 $\hat{H}$ via OLS dyadic variogram | Bands = 95% cross-path CI

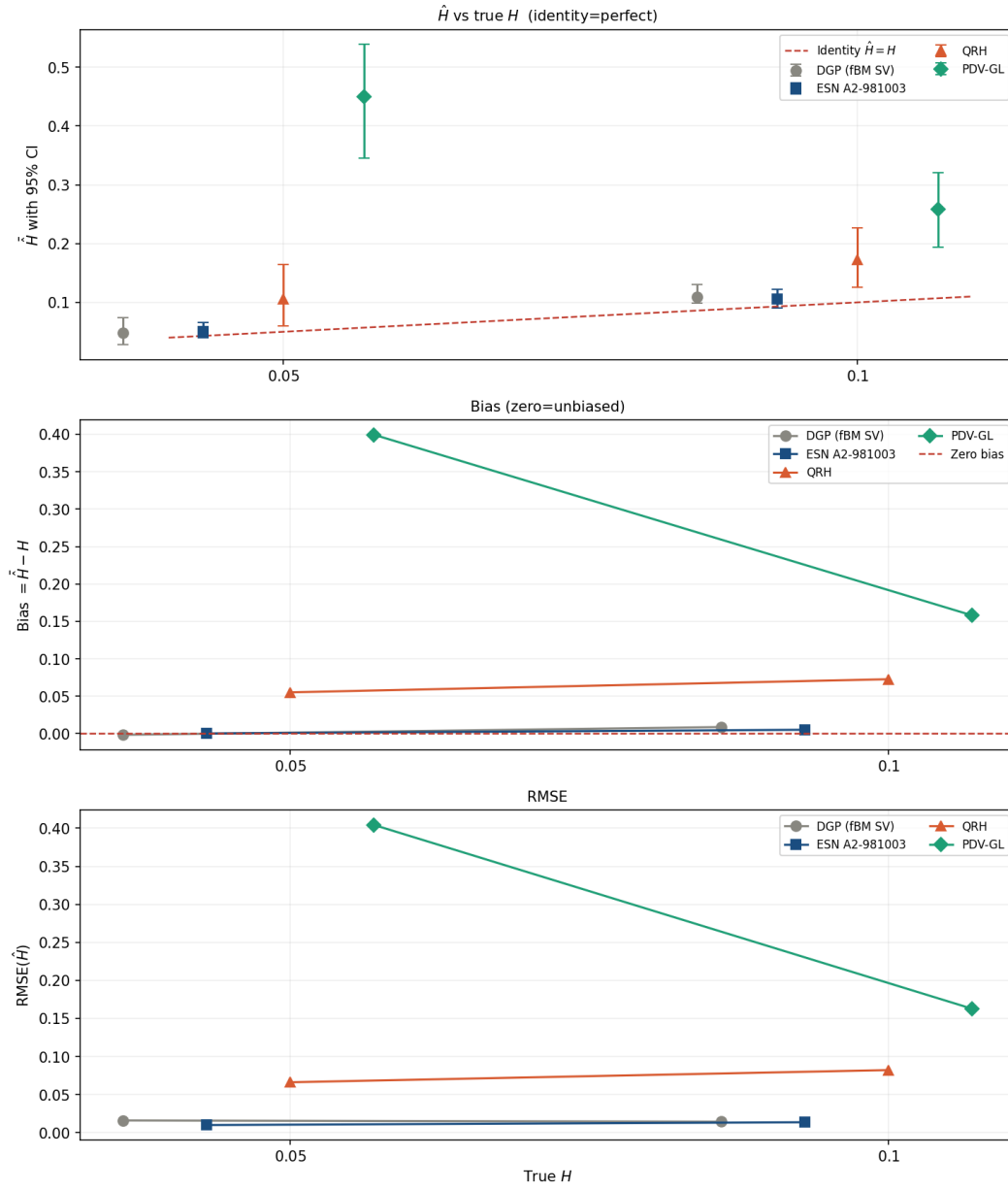


Figure 4: $\hat{H}$ vs. true H (top; identity line = perfect), bias (middle), and RMSE (bottom), per model, across both true H values tested at $\lambda = 1.0$.

References

- [1] J. Gatheral, T. Jaisson, M. Rosenbaum. *Volatility is rough*. QF 18(6), 2018.
- [2] F. Bourgey, J. Gatheral. *Quadratic Rough Heston*. SSRN:5239929, 2026.
- [3] J. Guyon, J. Lekeufack. *Volatility is (mostly) path-dependent*. QF 23(9), 2023.